sustainability

MDPI

Article

Effect of a Corn Silage-Based Finishing Diet on Growth, Carcass Composition, Meat Quality, Methane Emissions and Carbon Footprint of Crossbred Angus Young Bulls

Diana M. Soares [1,2,3,4,5], Sílvia Bernardino [6], Nuno Rodrigues [4], Ivo Gama [4], João M. Almeida [2,3,7], Ricardo F. M. Teixeira [5], José Santos-Silva [2,3,7], Susana P. Alves [1,2,3], Tiago Domingos [5,*], Cecile Martin [8], Gonçalo M. Marques [4,5] and Rui J. B. Bessa [1,2,3,*]

1 Faculdade de Medicina Veterinária, Universidade de Lisboa, Av. da Universidade Técnica, 1300-477 Lisboa, Portugal; diana.soares@terraprima.pt (D.M.S.)
2 CIISA—Centro de Investigação Interdisciplinar em Sanidade Animal, Av. da Universidade Técnica, 1300-477 Lisboa, Portugal
3 Laboratório Associado para Ciência Animal e Veterinária (AL4AnimalS), Av. da Universidade Técnica, 1300-477 Lisboa, Portugal
4 Terraprima-Serviços Ambientais, Centro de Negócios do Porto Alto, Avenida das Nações Unidas, No. 97, 2135-199 Samora Correia, Portugal
5 MARETEC—Marine, Environment and Technology Centre, Laboratory of Robotics and Engineering Systems (LARSyS), Instituto Superior Técnico, Universidade de Lisboa, Av. Rovisco Pais, 1, 1049-001 Lisboa, Portugal
6 Terraprima—Sociedade Agrícola, Quinta da França, 6250-111 Caria, Portugal
7 Instituto Nacional de Investigação Agrária e Veterinária (INIAV), Polo de Investigação de Santarém, 2005-424 Santarém, Portugal
8 Institut National de Recherche pour L'agriculture, L'alimentation et L'environnement (INRAE), VetAgro Sup, UMR Herbivores, Université Clermont Auvergne, F-63122 Saint-Genès-Champanelle, France
* Correspondence: tdomingos@tecnico.ulisboa.pt (T.D.); rjbbessa@fmv.ulisboa.pt (R.J.B.B.)

check for updates

Academic Editor: Jan Hopmans

Received: 31 July 2025
Revised: 15 September 2025
Accepted: 17 September 2025
Published: 19 September 2025

Citation: Soares, D.M.; Bernardino, S.; Rodrigues, N.; Gama, I.; Almeida, J.M.; Teixeira, R.F.M.; Santos-Silva, J.; Alves, S.P.; Domingos, T.; Martin, C.; et al. Effect of a Corn Silage-Based Finishing Diet on Growth, Carcass Composition, Meat Quality, Methane Emissions and Carbon Footprint of Crossbred Angus Young Bulls. *Sustainability* **2025**, *17*, 8417. https://doi.org/10.3390/su17188417

Correction Statement: This article has been republished with a minor change. The change does not affect the scientific content of the article and further details are available within the backmatter of the website version of this article.

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

Abstract

Using locally produced forage and agro-industrial by-products can reduce dependence on imported feed and competition for human food sources, while improving meat quality. However, the overall effect of this feeding strategy on global greenhouse gas emissions must be evaluated to provide a comprehensive assessment of sustainability. This study aimed to test whether replacing the conventional concentrate finishing diet with a total mixed ration (TMR) diet based on maize silage and brewer's spent grains (BSG) would improve meat quality without compromising productive performance, carcass composition, and the carbon footprint (CFp) of finishing beef cattle. Twenty crossbred young bulls were randomly distributed among 4 pens and randomly allocated to 2 treatments: Control—a conventional diet based on commercial concentrate and wheat straw or TMR—a maize silage-based diet with BSG, concentrate, and straw. Dry matter intake and average daily gain were 13% and 15%, respectively, lower in the TMR treatment than in the Control treatment. Daily methane emissions were 59% higher in the TMR treatment. However, life cycle assessment results revealed no differences in the CFp, and the beef from TMR treatment achieved higher meat quality. In conclusion, a maize silage-based diet offers a cost-effective alternative to conventional diets, with a lower environmental impact and improved beef quality.

Keywords: total mixed ration; beef cattle; carcass composition; meat fatty acids; environmental footprint

1. Introduction

Beef production is a major industry worldwide that has been under strong social pressure due to its significant contribution to the emissions of greenhouse gases (GHG), particularly methane (CH_4), produced in the cattle digestive tract [1]. The beef sector is estimated to account for between 2.2% and 6% of global anthropogenic GHG emissions [2,3]. Nevertheless, the carbon footprint (CFp) varies largely with the type and intensification level of beef production systems [4]. The most effective strategies to reduce CH_4 emissions involve optimizing diets and incorporating feed additives [5,6]. In many beef production systems, calves remain with cows on natural or sown pasture until weaning and are slaughtered after a fattening period of variable duration [7]. During this finishing phase, nutrient-dense diets, primarily based on high-protein concentrates, are provided to maximize growth [7]. In Europe, including Portugal, the animal feed industry relies heavily on imported raw materials [8]. Including locally sourced raw feeds and using industrial food by-products in finishing diets can enhance sustainability by reducing reliance on imports and competition with human food. In this context, the inclusion of locally produced forage in finishing diets should be promoted [7]. Total mixed rations (TMR) diets, incorporating high-quality forages such as maize silage (MS), are becoming increasingly common. Incorporating MS into finishing diets allows for the reduction in concentrate use, which can lower production costs without compromising performance [9]. Compared to concentrate-based diets, the TMR forage-based diets can also improve the nutritional profile of beef by increasing n − 3 polyunsaturated fatty acids (PUFA), vaccenic (t11-18:1), and rumenic acids (c9,t11-18:2) while preventing the accumulation of undesirable trans fatty acids (FA) like t10-18:1 [7]. Previous studies suggest that TMR with MS can also improve meat sensory quality traits [10]. An advantage of using TMR diets is the facility to include locally produced agro-industrial by-products unsuitable for human consumption in cattle finishing diets. However, their availability is often restricted seasonally and they have a higher fibre content, which may increase CH_4 emissions, hindering the efforts to reduce net GHG emissions [7]. Increasing the digestibility of diets does not always result in lower total GHG emissions, as other sources of emissions on the farm may increase [11]. To accurately assess the net impact of dietary changes on GHG emissions, it is essential to use a systemic approach like life cycle assessment (LCA) that accounts for GHG emissions from all stages of production, including feed production, manure management, and transport, providing a comprehensive picture of the total emissions from finishing beef production systems [12]. Thus, the net effect on GHG emissions of reducing beef cattle finishing diets' cereals and oilseeds meals by incorporating maize silage and agro-industrial by-products, such as brewers' spent grain, needs to be evaluated. Therefore, the aim of this study was to test the hypothesis that replacing the conventional concentrate-straw finishing diet with a TMR diet based on MS and brewers' spent grains would improve meat quality without compromising productive performance, carcass composition, CH_4 emissions, and CFp of finishing beef cattle.

2. Materials and Methods

All procedures were conducted following the European Union's rules for the use of animals in experiments (Directive 2010/63/EU) [13]. This study was approved by the Research and Education Ethics Committee (CEIE) of the Faculty of Veterinary Medicine, University of Lisbon, Portugal (Proc. 038/2023).

2.1. Animals, Diets, and Experimental Design

Twenty young crossbred males, sired by Angus bulls and Limousin or Charolais cows and born between September and November 2021, were selected for this experiment.

The young males had an average age of 13.6 months and an average live weight (LW) of 473 ± 37.6 kg. The experiment was conducted in a farm (Quinta da França, Covilhã, Portugal, 40°16′24.73″ N, 7°25′55.535″ W) where 4 pens (5.8 m × 10 m), along a 4 m wide corridor and a 2 m wide rear sleeve that directed the animals to the 50 m^2 outdoor weighing area, were used for this trial. The pens had a concrete floor and an automatic stainless steel water distribution system. On arrival, the young bulls were blocked by weight and farm of origin and randomly distributed among the four pens, five animals per pen. The trial began after 12 days of adaptation to the diets and environmental conditions. After the adaptation period, the LW was 500 ± 38.5 kg (mean ± sd).

The pens (experimental units) were randomly allocated to two dietary treatments: Control—a conventional diet based on commercial concentrate (NANTA II Nutrição S.A., Ovar, Portugal) and wheat straw; TMR—a forage-based diet consisting of maize silage, brewer's spent grains, concentrate, and the same straw. The TMR was prepared daily with a vertical mixer (Mammut, Profi Mix, Gurten, Austria). Maize silage was produced on the farm in 2022 and the brewer's spent grains were produced in November 2022 and supplied by a national brewer. The composition of the diets used is shown in Table 1. Weekly, sub-samples of the Control and TMR diets were collected and frozen. The sub-samples were pooled monthly, and the composite samples were analysed for the chemical composition, which is shown in Table 2.

Table 1. Chemical composition of diet ingredients.

	Ingredient					
Chemical Composition (% DM [1])	**Control Concentrate [2]**	**TMR Concentrate [3]**	**GF Pelleted [4]**	**Maize Silage**	**Breer's Spent Wet Grains**	**Straw**
Dry Matter (%)	88	88.3	89	29.8	29.7	97.5
Crude Protein	14.7	13.7	16.9	7.73	29.0	-
Ether Extract	7.3	6.9	5.6	-	-	-
Crude Fibre	6.6	5.0	6	22.5	20.8	-
Starch	42.6	58.6	45	33.0	1.2	-
NDF [5]	22.5	14.2	20.2	46.5	43.2	-
ADF [6]	8.2	6.0	7.9	26.0	27.4	-
Calcium	0.74	1.6	1.1	-	-	-
Phosphorus	0.65	0.53	0.45	-	-	-
Sodium	0.20	0.43	0.45	-	-	-

[1]—Dry matter; [2]—Ground concentrate used in the control treatment; [3]—Ground concentrate incorporated in the TMR diet; [4]—Concentrate pellets used as bait in the Greenfeed apparatus; [5]—Neutral detergent fibre; [6]—Acid detergent fibre.

Table 2. Ingredients, chemical composition, and fatty acid (FA) profile of the diets.

	Diets	
	Control	**TMR**
Ingredients (g/kg DM)		
Control concentrate	887	-
TMR concentrate	-	328
Maize silage	-	471
Brewer's spent wet grains	-	189
Straw	113	12
Chemical composition (g/kg DM)		
DM (g/kg as fed)	862	393
Crude Protein	132	137

Table 2. *Cont.*

	Diets	
	Control	**TMR**
Ether Extract	75	51
Crude Fibre	91	162
NDF [1]	321	491
ADF [2]	123	235
ADL [3]	21	30
Sugar	34.1	18.5
Starch	396	312
Ash	54	76
Calcium (mg/g)	4.79	4.90
In Vitro Digestibility (g/kg)		
DM [4]	789	694
OM [5]	779	709
Metabolizable Energy [6] (MJ/kg DM)	11.9	10.9
Fatty acid profile (g/kg FA)		
14:0	-	0.9
16:0	300	281
18:0	60	36
c9-18:1	223	200
c11-18:1	5.4	7.5
18:2n − 6	348	416
20:0	17.9	7.6
18:3n − 3	15.3	38.5
22:0	12.0	5.6
24:0	18.5	7.7
Total FA (g/kg DM)	65.6	39.5
Feed cost (€/kg DM)	0.441	0.305

[1]—Neutral detergent fibre; [2]—Acid detergent fibre; [3]—Lignin; [4]—Dry matter; [5]—Organic matter; [6]—Calculated using the formula by [14].

Animals were fed ad libitum, once daily, at approximately 8:00 a.m., allowing for a 10% refusal rate. Orts in each pen were recorded daily. Animals were weighed every 2 weeks to quantify growth performance. The trial lasted for 12 weeks, after which the animals were slaughtered as described in point 2.3. Feed costs were computed using raw materials market prices in 2022.

2.2. Gas Emissions Measurement

Methane and carbon dioxide (CO_2) emissions of animals were recorded using GreenFeed equipment (GF) (GreenFeed—Large Animals, C-Lock Inc., Rapid City, SD, USA). The pelleted compound feed (Rações Zêzere S.A., Ferreira do Zêzere, Portugal) used as bait in the GF is described in Table 1.

The equipment was automatically calibrated every three days for CH_4 (zero and span) using the manufacturer's built-in automatic calibration system. According to the manufacturer's instructions and previously described [7]. Every 15 days, the air filter was changed and cleaned. The GF was available in each pen with free access for seven days. After this time, the GF equipment was moved to the subsequent pen. So, throughout the experimental period, the animals in each pen had access to the GF unit for three 7-day periods separated by 4 weeks. The GF was instructed to deliver eight doses of bait a maximum of five times per day, with no minimum time between each visit. Each dose weighed an average of 38 g. The homogeneity of the animals' ages and LW attempts to

prevent social dominance during visits to the GF. Metallic alley barriers were also installed at the entrance to the GF to ensure that only one animal was visited at a time.

2.3. Slaughter and Meat Sample Collection

The animals were transported and slaughtered at a commercial abattoir (OVIGER—P.T.C. Carnes e Derivados S.A., Castelo Branco, Portugal) located 48 km from the farm.

Animals were stunned using a penetrating captive bolt and slaughtered by exsanguination following the European standards for the protection of animals at the time of killing [15]. After weighing, carcass conformation and fattening were evaluated after dressing using the SEUROP grading scheme [16]. The carcasses were split in half along the vertebral column and chilled in a cold room at 0–4 °C. At 48 h post-slaughter, longissimus lumborum (LL) muscle samples and the "Rib Primal 103" [17] portion were taken from the left half-carcasses. The 9th, 10th, and 11th ribs were obtained from the "Rib Primal 103" using a saw [18]. The sections were vacuum-packed and frozen at −20 °C until they were dissected.

2.4. Prediction of Carcass Composition and Carcass Weight, Energy, and Protein Gain

The 9th, 10th, and 11th rib cuts were weighed and dissected to obtain the lean, fat, and bone weights to allow the prediction of the carcass proportions of adipose tissue, lean, and bone using the equations proposed by [18]:

$$\text{Proportion of fat} = 3.06 + 0.82X \quad (1)$$

$$\text{Proportion of lean} = 15.56 + 0.81X \quad (2)$$

$$\text{Proportion of bone} = 4.30 + 0.61X \quad (3)$$

where X is the percentage of the respective separable tissue in the rib cut.

The fat and lean tissues from each carcass were ground three times in a 9.5 mm meat grinder (Braher Internacional, S.A., Andoain, Spain) and a homogeneous representative sub-sample was taken.

The proportions of lean body mass (LBM), protein, and water in the carcasses were calculated according to the equations of the Agricultural Research Council [19]:

$$\text{LBM} = \text{Carcass weight} - \text{Fat weight in carcass} \quad (4)$$

$$\text{Protein} = \text{LBM} \times 0.216 \quad (5)$$

$$\text{Water} = \text{LBM} \times 0.729 \quad (6)$$

The energy content of the carcasses was estimated by the sum of the energy value of protein (23.6 kJ/g) and the energy value of fat (39.3 kJ/g) in the carcass. The carcass weight (CW) gain and carcass energy gain were estimated using linear regression models.

A linear regression model was fitted to estimate carcass energy (CE) content from CW yielding. Carcass protein (CPr) gain was estimated through the slope of CW gain derived from a linear regression. These equations are presented in sub-chapter Section 3.2.

2.5. Analytical Procedures

2.5.1. Feeds

The diets used in the experiment were analysed for dry matter (DM), crude protein, ether extract, crude fibre, ash and minerals as described by Santos-Silva et al. [7]. Starch and sugar were determined according to the method described by Clegg [20]. Neutral detergent

fibre (NDF), acid detergent fibre (ADF) and acid detergent lignin (ADL) were determined according to the method [21]. The in vitro digestibility was determined according to [22]. Metabolizable energy (ME) was calculated using the formula by [14]. Direct transesterification of dietary lipids was adapted according to the method from [23] using nonadecanoic acid (19:0) as the internal standard and fatty acid methyl ester (FAME) analysed by gas chromatography.

2.5.2. Muscles, Subcutaneous and Inter-Muscular Adipose Tissue

For chemical analysis and pH determination, 200 g of LL was taken from the 9th, 10th, and 11th rib cuts, stripped of visible adipose and connective tissues, minced in a Moulinex food processor (Group SEB Portugal Lda, Lisbon, Portugal) for 3 × 5 s, then vacuum-packed and stored at −20 °C. The analyses were performed in duplicate. Briefly, dry matter was determined according to [24,25], ether extract content was determined by the Soxhlet method [26], using a Soxhlet extractor (VELP Scientific SER 148/6 Solvent Extractor, Usmate, Italy) and pH at 1:10 in 0.1 M KCl solution according to [27].

Intramuscular lipids of freeze-dried LL samples were extracted according to the method of [28], but where chloroform was replaced by dichloromethane. The extracted intramuscular lipids and intermuscular fat (ImF) samples taken from the 9th, 10th, and 11th rib cuts were transesterified according to the methods detailed in [29]. For FAME identification, commercial standard mixtures (FAME mix containing 37 components, supplied by Supelco Analytical Products, Darmstadt, Germany) and published chromatograms [30,31] were used to compare FAME retention times. Additional FAME identification was performed when necessary, using electron impact mass spectrometry with a Shimadzu GC-MS QP2010 Plus (Shimadzu, Kyoto, Japan).

The colour of subcutaneous fat (ScF), ImF, and LL meat was assessed seven days post-mortem using a CR-400 chroma meter (Konica Minolta Inc., Tokyo, Japan) set up with a 10 mm diameter aperture, a D65 illuminant, and a 2° observer. The chroma meter was calibrated daily using a white standard plate (D65:Y84.9, x:0.3199, y:0.3359). Measurements were recorded in the CIELAB system, where L* is lightness, a* is redness, and b* is yellowness. Colour saturation (C*) and hue angle (H*) were calculated as $(a^{*2} + b^{*2})^{1/2}$ and $\tan^{-1}(b^*/a^*) \times (180/\pi)$, respectively [32].

Meat cooking loss and Warner–Bratzler shear force were determined as described in detail by Santos-Siva et al. [7].

2.6. Sensory Panel

The sensory analysis was carried out by a panel of 9 members from INIAV-Fonte Boa, selected and trained according to [33], who assessed the (1) tenderness, (2) juiciness, (3) flavour, and (4) overall perception of the meat in seven sessions. For each session, six frozen LL samples were randomly selected: three from each diet, with the same ageing time, allowed to thaw for 24 h at 2 °C, and prepared as described by [7]. For the sensory analysis, a defined scale from 1 to 6 was used, with values between: 1—extremely tough, dry, soft, or unacceptable; and 6—extremely tender, juicy, and intense, or acceptable.

2.7. Carbon Footprint Calculation

A partial life cycle assessment (LCA) approach was used to account for additional direct and indirect emissions in addition to the measurements of CH_4 and CO_2. The LCA is a method that considers all direct and indirect potential environmental impacts that are due to a given production system [34]. As it only uses global warming potential as an impact indicator, the LCA approach used here can be considered a CFp metric. In the CFp assessment, the functional units used were 1 kg live weight (LW) gain, 1 kg carcass weight (CW) gain, and 1 kg of CPr gain, all of which occurred during the finishing period only. The

functional unit used was cradle to farm gate, but the temporal boundaries of the analysis included only the finishing period of the young bulls. Therefore, the pre-fattening period of the animals, the transport, slaughter process, carcass deboning, and meat processing up to the point of consumption were not included in the LCA calculations. This choice was made because those stages are not affected by the treatments studied here. The CFp was calculated individually for each animal using the online tool developed by Terraprima (https://www.terraprima.pt/en/area-de-actividade/10 (accessed on 5 January 2025)). This tool calculates the contribution to climate change measured in kg CO_2eq using a classification model based on the IPCC 4th Assessment Report [35]. The CFp calculation included emissions from rumen digestive fermentation, manure in the pens, and emissions from the life cycle of feed ingredients production. The GHG emissions from the digestive tract were acquired using the GF measurements described previously. The emissions from excrement during confinement were estimated, individually for each bull, by using equation 10.24 (Tier 2) from [36], refined in 2019, considering specifically the duration of the trial, the gross energy intake, and the digestibility of the diets. For maize silage, the emission factor was calculated using field data for fertilizer applications and operations in the farm where the trial took place, applying emission factors from [35,37]. For both concentrate feeds and straw, the emission factors were calculated using the Ecoalim V1 database from Agribalyse [38]. For brewer's grains, we considered an average of beer production emission factors present in the CarbonCloud Database [39] of 0.676 kg CO_2eq/L of beer. Then we considered that for each 100 L of beer, 20 kg of brewer's grains are produced and made an economic allocation, using 2.15 €/L of beer and 0.08 €/kg of brewer's grains as the economic value, in order to calculate the brewer's grains final emission factor. The resulting emission factors were 0.037 kg CO_2eq/kg of fresh weight silage, 0.045 kg CO_2eq/kg of fresh weight straw, and 0.025 kg CO_2eq/kg of fresh brewer's grains. For the concentrate feeds in the Control diet, the calculated emission factor was 0.574 kg CO_2eq/kg and 0.542 kg CO_2eq/kg in the TMR diet.

2.8. Statistical Analysis

Statistical analysis was performed using Proc MIXED (SAS Institute Inc., Cary, NC, USA). The variables were analysed considering the pen as the experimental unit and the diet as the single fixed effect, after checking data for homogeneity of variance. The animals within the pens were treated as subsampling using the statement "repeated/sub = pen(diet) type = cs". The meat sensory data were analysed with a similar model but including the panellist as a block. Average daily gain was obtained through the slope of the random regression mixed model of the LW at several time points, using the ARH(1) covariance structures for the repeated measurements within animals and the pen(treatment) as a random factor. In all cases, the covariance structure (CS and ARH(1)) was selected according to the model fit statistics.

A linear regression between final live weight (FLW) and CW was fitted (CW = 0.618 kg $\pm$ 0.038 $\times$ FLW $-$ 38.48 kg, R^2 = 0.935, RSD = 8.88, n = 20) and used to estimate the carcass weight gain using the initial live weight. Carcass composition and energy content data were computed as explained in Section 2.5.

3. Results

3.1. Feed Intake, Growth Performance, and Carcass Traits

The feed intake, growth performance, carcass traits, and feeding costs during the finishing period are presented in Table 3. The average daily DM intake (DMI) was 13.2% lower in TMR (p = 0.035) compared to the Control group. Consistently, the average daily gain (ADG) was significantly higher in Control, with a value of 1693 g/day, compared to 1442 g/day in

the TMR treatment (p = 0.024). Despite that, the difference of 22 kg of LW gain observed between treatments did not reach statistical significance (p = 0.19). No statistical differences were detected for the initial weight (p = 0.92), slaughter weight (p = 0.41), cold carcass weight (p = 0.41), dressing (p = 0.76), and feed efficiency (p = 0.72).

Table 3. Effects of the diet on growth performance, carcass characteristics, and feed costs of crossbred Angus young bulls.

	Diets		SEM [1]	*p*-Value
	Control	TMR		
DM [2] intake (kg/d)	10.6	9.2	0.191	0.035
ME [3] intake (MJ/d)	126	100	0.38	0.001
Initial LW [4]	501	499	12.5	0.920
Slaughter LW (kg)	634	610	12.5	0.302
LW gain (kg)	133	111	8.1	0.191
Average daily gain (g/d)	1693	1442	78	0.024
Cold carcass weight (kg)	354	338	9.0	0.345
Dressing percentage (%)	55.7	55.4	0.513	0.757
Feed efficiency	0.16	0.15	0.009	0.717
Feed costs (€/d)	4.67	2.80	0.072	0.003
Feed costs (€/kg LW gain)	2.86	2.05	0.137	0.053

[1]—Standard error of the means; [2]—Dry matter; [3]—Metabolizable energy; [4]—Live weight.

The feed costs per day were approximately 40% lower in TMR than in Control (p = 0.003) and the feed cost per kg LW gain was still 28% lower in TMR than in Control, although not reaching statistical significance (p = 0.054).

3.2. Carcass Grading and Composition

The results of carcass classification according to the SEUROP system showed that in the TMR treatment, 9 out of 10 animals were classified as "R" (Good) and only one carcass was classified as "U" (Very good). In the Control treatment, 50% of the carcasses were classified as 'R' and 50% as 'U'. Regarding the level of fat cover, all the carcasses were classified as class 2 (Slight) in both treatments.

The carcass composition results are presented in Table 4. The proportion of separable fat, lean, and bone in the rib cuts was not significantly different between the diets (p > 0.5) and averaged 21.9%, 57.8%, and 17.6%, respectively. This resulted in estimates for the separable fat, lean, and bone in the carcass of 21.0%, 62.4%, and 15.0%, respectively. Applying this estimation to CW and using the mean chemical composition of lean and fat, we estimated that the carcass had an average of 273 kg of lean body mass, 59 kg of protein, 61 kg of bone, 199 kg of water, and 72.4 kg of fat, none differing significantly among treatments.

Table 4. Effects of the diet on physical composition of ninth-tenth-eleventh-rib cuts and on total % separable fat, separable lean and separable bone, chemical constituents, and energy content (MJ) in carcasses of crossbred Angus young bulls.

	Diets		SEM	*p*-Value
	Control	TMR		
9th–11th Ribs Dissection [1] (%)				
Separable fat	22.3	21.4	2.16	0.795
Separable lean	57.8	57.7	1.75	0.987
Separable bone	17.3	17.8	0.13	0.102

Table 4. *Cont.*

	Diets		SEM	*p*-Value
	Control	TMR		
Carcass estimations [2] (%)				
Separable fat	21.4	20.6	1.78	0.795
Separable lean	62.4	62.3	1.42	0.987
Separable bone	14.9	15.1	0.08	0.102
Carcass (kg)				
Weight	354	338	9.0	0.345
Lean Body Mass	278	268	12.9	0.638
Protein	60.1	57.9	2.79	0.638
Bone	61.2	60.4	1.65	0.776
Water	202	196	9.4	0.638
Fat	75.2	69.6	4.87	0.499
Carcass energy (MJ)	4375	4102	138	0.298
Carcass weight (CW) gain [3] (kg)	82.4	68.0	5.03	0.191
Carcass energy gain [3,4] (MJ)	973	809	59.7	0.191
Carcass protein (CPr) gain [3] (kg)	15.6	12.9	0.95	0.191

[1] Observed values. [2] Estimated by the equations generated by [18]; [3] Estimated values of predicted linear regression equations; [4] adjusted to a carcass fat percentage of 21%.

A linear regression model was fitted to estimate CE content from CW, yielding:

$$CE = (9.31 \pm 2.96)\ MJ \times CW + 1020 \quad (7)$$

with R^2 = 0.355, RSD = 437 and n = 20. However, this model showed poor fit due to the large variation in estimated carcass fat content (15.8% to 27.9%). Fitting a linear regression model with both CW and carcass fat (%) yielded a model with a well-fitted (R^2 = 0.995, RSD = 43 MJ) and a slope of 11.85 MJ per kg of carcass gain adjusted to carcass fat percentage of 21%. Diet and its interactions with regressors were tested and had no effects ($p > 0.79$) and were removed from the model. The value of 11.85 MJ/kg carcass was used to estimate the energy value of carcass gain. Protein gain was estimated through the slope 0.189 kg/kg of CW gain derived from a linear regression relating CPr to CW:

$$CPr = (0.189 \pm 0.019)\ kg \times CW - 6.43 \quad (8)$$

with R^2 = 0.850, RSD = 2.76 and n = 20.

The estimated gross energy content of carcass (4375 MJ in Control and 4102 MJ in TMR), carcass energy gain (973 MJ in Control vs. 809 MJ in TMR), and protein gain (15.6 kg in Control and 12.9 kg in TMR) were numerically higher for the Control than in the TMR treatment, but not significantly different (p = 0.191).

3.3. CH_4 and CO_2 Production

The results for CH_4 and CO_2 production for the control and TMR treatments assessed by GF equipment are reported in Table 5. All 20 animals visited the GF with valid emission measurements for all animals. The animals spent an average of 4 min 30 s in the GF during each visit and visited it between 1 and 7 times a day. The digestive CH_4 emissions were higher ($p < 0.05$) for TMR than for Control treatments, regardless of the unit as expressed. Daily CH_4 emissions were higher in the TMR group compared to the Control group (192 vs. 121 g/day, p = 0.02). Similar results were observed when CH_4 emissions are expressed per kg of DMI (21.0 g for TMR vs. 11.5 g for Control, p = 0.009) per kg of LW gain (150

g for TMR vs. 79 g for Control, $p = 0.008$) and per kg of CPr gain (64.3 kg for TMR vs. 33.3 kg for Control, $p = 0.008$). These daily CH_4 emissions correspond to a loss of 9.65 MJ/d (5.7% of GE intake) for the TMR-fed animals and of 6.08 MJ/d (3.0% of GE intake) for the Control-fed animals.

Table 5. Effect of diet on digestive methane (CH_4) and carbon dioxide (CO_2) production and energy loss estimation during finishing period (12 weeks) of crossed Angus young bulls.

	Diets		SEM	*p*-Value
	Control	TMR		
CH_4 (g/d)	121	192	7.2	0.020
CH_4 (g/kg DMI)	11.5	21.0	0.64	0.009
CH_4 (g/kg LW gain)	78	150	4.66	0.008
CH_4 (g/kg CW gain)	126	243	7.54	0.008
CH_4 (kg/kg CPr gain)	33.3	64.3	1.99	0.008
CH_4 Energy losses				
CH_4 (MJ/d)	6.08	9.65	0.361	0.020
CH_4 (% GE Intake)	3.0	5.7	0.20	0.011
CO_2 (kg/d)	8.94	8.88	0.05	0.511
CO_2 (kg/kg DMI)	0.84	0.97	0.018	0.039
CO_2 (kg/kg LW gain)	5.72	7.07	0.498	0.196
CH_4/CO_2 (g/kg)	13.6	21.7	0.34	0.004

The CO_2 production values were similar for the two treatments when represented as kg/day, averaging 8.91 kg/d. The same was observed when CO_2 emissions were expressed as kg/kg LW gain (5.72 in Control and 7.07 in TMR, $p = 0.196$), but when expressed per kg of DMI, they were higher ($p = 0.039$) in TMR (0.97 kg/kg DMI) when compared to the Control treatment (0.84 kg/kg DMI). The CH_4/CO_2, molar ratio (g/kg) was larger ($p = 0.004$) in TMR (21.7 g/kg) than for the Control (13.6 g/kg) treatment.

3.4. Meat Quality

The effects of diet on the chemical, physical, and sensory traits of the LL muscle evaluated at 7 days post-mortem are presented in Table 6. There were no significant effects of treatments on muscle DM, intramuscular fat, pH, cooking losses, and shear force ($p > 0.05$). There were no significant effects of diet on the meat colour parameters ($p > 0.05$), except for the L* parameter that was higher for TMR than for the Control treatment (42.5 vs. 40.2, $p = 0.046$). The colour parameters of ScF and ImF did not differ between treatments. The colour parameters averaged L*, 75.5 ± 0.71; a*, 2.46 ± 0.683; b*, 8.61 ± 0.713 for ScF and L*, 80.5 ± 0.47; a*, 0.32 ± 0.224; b* 7.84 ± 0.250 for ImF.

Table 6. Effects of the diet on chemical composition and on physical and sensorial properties of *m. Longissimus lumborum* of crossbred Angus young bulls.

	Diets		SEM [1]	*p*-Value
	Control	TMR		
Dry matter (%)	27.6	26.7	0.26	0.128
Fat (%)	3.08	2.88	0.186	0.524
pH	5.75	5.73	0.032	0.657
Cooking loss (%)	32.7	31.7	0.42	0.236
Shear force (N/cm^2)	57.1	45.2	4.22	0.186

Table 6. *Cont.*

	Diets		SEM [1]	*p*-Value
	Control	TMR		
Colour parameters				
*L**	40.2	42.5	0.36	0.046
*a**	23.0	24.7	0.79	0.272
*b**	13.6	15.1	0.48	0.165
*C**	26.7	28.8	0.90	0.231
*H**	30.5	31.4	0.31	0.167
Sensory traits				
Tenderness	3.48	4.16	0.107	0.046
Juiciness	3.03	3.45	0.060	0.039
Flavour	2.43	2.49	0.044	0.484
Overall acceptability	3.84	4.27	0.082	0.065

[1]—Standard error of the means; *L**—lightness; *a**—redness; *b**—yellowness; *C**—chroma; *H**—hue.

The tasting panel graded the meat from TMR-fed animals as being more tender (4.16 vs. 3.48, $p = 0.045$) and with higher juiciness (3.45 vs. 3.03, $p = 0.039$) than that from Control, but with similar flavour (≈2.46, $p = 0.484$). The overall acceptability of the meat tended ($p = 0.065$) to be higher for TMR treatment (4.27) than for Control (3.84).

3.5. Fatty Acid Composition of Lean Meat and Adipose Tissue

The FA composition of LL expressed as mg/100 g of meat is presented in Table 7 and detailed in Table S1 of the Supplementary Material. The total FA concentration was similar in Control and in TMR (2273 vs. 1899 mg FA/100 g meat, $p = 0.188$). Most of the FA, including the most abundant (i.e., 16:0, 18:0, and c9-18:1), did not present differences between treatments. However, despite the lack of statistical difference observed for c9-18:1, the difference between means is large (559 mg vs. 397 mg), and the variability of values is also large (Coefficient of variation = 78%). The clear differences observed between diets were restricted to the 18:2n − 6 and several biohydrogenation intermediates (BHI). The 18:2n − 6 proportion in meat was 49% higher in the Control than in the TMR treatments. Moreover, the 20:4n − 6 and 20:5n − 3 also tended ($p < 0.1$) to be higher in the Control than in the TMR treatment. Several BHI were present in higher proportions (i.e., the coeluted peak of t6-18:1, t7-18:1, and t8-18:1, the t10-18:1, t12-18:1, c13-18:1, and c15-18:1; $p < 0.05$) or tended to be higher (i.e., t9-18:1, c12-18:1, and the coeluted peak of c14-18:1 and t16-18:1; $p < 0.1$) in the Control than in the TMR treatment.

Table 7. Effects of the diet on fatty acid (mg/100 g of meat) of *m. Longissimus lumborum* (LL) of crossbred Angus young bulls.

	Diet		SEM	*p* Value
	Control	TMR		
Total FA	2273	1899	134	0.188
FA profile				
16:0	698	678	24.5	0.617
*c*9-16:1 [1]	42.6	30.7	6.08	0.299
18:0	534	522	10.7	0.499
*c*9-18:1	559	397	64.6	0.219
18:2n − 6	114	58	7.4	0.034
18:3n − 3	5.1	3.6	0.68	0.265
20:4n − 6	17.6	13.2	0.78	0.058
20:5n − 3	1.7	1.0	0.15	0.083

Table 7. *Cont.*

	Diet		SEM	*p* Value
	Control	TMR		
Biohydrogenation intermediates (BHI)				
*t*10–18:1	34.0	8.6	3.50	0.036
*t*11–18:1	17.8	13.2	1.23	0.118
*c*9,*t*11-18:2	5.6	3.5	0.63	0.150
Partial sums				
SFA	1327	1272	37.7	0.410
MUFA	745	508	82.3	0.179
cis-MUFA	659	463	76.2	0.211
trans-MUFA	86.4	45.3	6.33	0.044
PUFA	147	83	9.87	0.045
n − 6 PUFA [2]	138	76	8.9	0.039
n − 3 PUFA [3]	8.9	7.0	0.97	0.313
BHI [4]	117	63	9.9	0.061
FA Ratios				
n − 6/n − 3 PUFA	18.6	11.5	0.81	0.025
SCD-17 [5]	22.6	19.1	1.45	0.236
*t*10-/*t*11-18:1	1.8	0.6	0.03	0.001

[1] Coelutes with minor amounts of anteiso-17:0; [2] n − 6 PUFA = 18:2n − 6 + 18:3n − 6 + 20:2n − 6 + 20:3n − 6 + 20:4n − 6 + 22:4n − 6; [3] n − 3 PUFA = 18:3n − 3 + 20:5n − 3 + 22:5n − 3; [4] BHI = Sum of biohydrogenation intermediates presented individually in the table; [5] SCD17, Stearoyl-CoA activity index computed as c9-17:1/(c9-17:1 + 17:0) × 100; Values with different superscripts are significantly different ($p < 0.05$).

The individual FA results are translated into the partial sums of FA, where only the trans-MUFA, PUFA, and n − 6 PUFA are higher ($p < 0.05$), and the BHI tended ($p = 0.061$) to be higher in the Control than in the TMR treatment. The n − 6/n − 3 PUFA ratio was 38% lower ($p = 0.025$) in the TMR than in the Control treatment. The t10-18:1/t11-18:1 ratio for Control meat was 200% higher ($p = 0.001$) than that of the TMR meat. No differences were observed for the SCD-17 ($p = 0.236$), a proxy of stearoyl-CoA activity.

The FA profile of intermuscular adipose tissue, expressed as mg/g of total FA, is presented in Table 8 and detailed in Table S2 of the Supplementary Material. Most of the FA were affected by treatments. Compared to Control the TMR treatment increased iso-15:0, anteiso-15:0, 16:0, 17:0, and 18:0, and decreased the c9-17:1, c9-18:1, c11-18:1, 18:2n − 6, c9-20:1, 20:2n − 6, and most of BHI, notably the t10-18:1 and c9,t11-/t7,c9-18:2. The 18:3n − 3 and the t11-18:1 did not differ between treatments.

Table 8. Effects of the diet on fatty acid (FA) (mg/g of total FA) of intermuscular adipose tissue from *m. Longissimus lumborum* (LL) of crossbred Angus young bulls.

	Diet		SEM	*p* Value
	Control	TMR		
Total FA (mg/g DM)	863	815	29.4	0.366
FA profile				
16:0	265	307	1.5	0.002
*c*9–16:1 [1]	26.0	26.4	0.42	0.530
18:0	210	224	2.4	0.051
*c*9-18:1	312	287	2.4	0.017
18:2n − 6	29.4	18.8	0.14	0.001
18:3n − 3	1.4	1.3	0.04	0.137

Table 8. *Cont.*

	Diet		SEM	*p* Value
	Control	TMR		
Biohydrogenation intermediates (BHI)				
*t*10–18:1	28.0	12.1	1.03	0.008
*t*11–18:1	9.8	11.3	0.46	0.144
*c*9,*t*11-18:2	4.3	3.5	0.102	0.029
Other CLA [2]	0.6	0.2	0.05	0.030
Partial Sums				
SFA	532	589	1.38	0.001
MUFA				
cis-MUFA	369	340	2.4	0.013
trans-MUFA	51.5	35.8	0.99	0.008
PUFA	31.3	20.4	0.18	0.001
BHI	62.0	44.0	0.93	0.005
FA ratios				
n − 6/n − 3 PUFA	20.1	14.8	0.43	0.009
SCD-17 [3]	30.5	27.6	0.34	0.027
*t*10-/*t*11-18:1	3.16	1.12	0.153	0.011

[1] Coelutes with minor amounts of anteiso-17:0; [2] Sum of *t*10,*c*12-18:2 + *t*11,*c*13-18:2 and other minor *t*,*t* conjugated isomers of linoleic acid; [3] SCD-17, Stearoyl-CoA activity index computed as *c*9-17:1/(*c*9-17:1 + 17:0) × 100; Values with different superscripts are significantly different ($p < 0.05$).

These effects of individual FA are reflected in the higher concentration of SFA (589 vs. 532, $p = 0.001$) and lower concentration of cis-MUFA (369 vs. 340, $p = 0.013$) and trans-MUFA (52 vs. 36, $p = 0.008$), PUFA (31 vs. 20, $p = 0.001$), and BHI (62 vs. 44, $p = 0.005$) in the intermuscular adipose tissue of TMR-fed animals. The n − 6/n − 3 PUFA, SCD-17, and t10-18:1/t11-18:1 ratios are higher in the Control than in the TMR treatment.

3.6. Carbon Footprint

The effects of diet on various emission components of the CFp of the finishing period of young bulls are presented in Table 9. When the various components of the footprint are expressed in kg of CO_2eq per kg of LW gain, the CFp associated with digestive tract fermentation was 48% higher in animals fed the TMR diet compared to the Control diet ($p = 0.026$). In contrast, emissions related to feed production were about 46% lower in the TMR diet compared to the Control diet ($p = 0.018$). For manure management, emissions tended to be higher ($p = 0.059$) in the TMR than in the Control. Despite these variations, the total CFp during the finishing period was similar between treatments, with values of 5.9 kg CO_2eq/kg LW gain for the Control and 6.0 kg CO_2eq/kg LW gain for the TMR treatment ($p = 0.770$). When the footprint used a functional unit of kg of carcass gain and kg of protein gain, the results were also the same for both treatments. The average CFp was 9.8 kg CO_2eq/kg CW gain and 51.6 kg CO_2eq/kg of CPr gain, for both treatments.

Table 9. Effects of the diet on emissions components of partial life cycle analysis for the finishing period (12 weeks) of crossbred Angus young bulls.

	Diets		SEM [1]	*p*-Value
	Control	TMR		
kg CO_2eq/kg LW [2] gain				
Digestive tract	1.86	3.60	0.202	0.026
Feeds	3.70	2.00	0.166	0.018
Manure	0.29	0.41	0.022	0.059

Table 9. *Cont.*

	Diets		SEM [1]	*p*-Value
	Control	TMR		
Finishing period carbon footprint kg CO_2eq/kg CW [3] gain	5.9	6.0	0.34	0.770
Finishing period carbon footprint kg CO_2eq/kg CPr [4] gain	9.5	10.0	0.80	0.716
Finishing period carbon footprint	50.3	52.8	4.21	0.716

[1]—Standard error of the means; [2]—Live weight; [3]—Carcass weight; [4]—Carcass protein.

4. Discussion

The main aim of the trial was to test the hypothesis that replacing the conventional beef finishing diets based heavily on cereal- and oilseeds-derived feedstuffs concentrates with a TMR finishing diet based on maize silage and brewer's spent wet grains (BSWG) would improve meat quality and reduce costs, without compromising the productive performance and the CFp of beef.

4.1. Productive Performance

The DM intake of the TMR diet was lower than that of the Control, despite the much larger intake as a fresh diet. The lower rumen digestibility and passage rate of forage-based diets, together with their bulkiness, might restrict the DM voluntary intake due to the intake regulatory mechanism linked to rumen fill [40]. Avilés et al. [10] and Cooke et al. [41] also observed higher DMI in Limousine and Retinta young bulls and in Charolais crossbred heifers, respectively, fed a conventional concentrate diet than those fed a maize silage-based TMR. On the other hand, the TMR diet contained 18.9% of DM as BSWG, which are difficult to preserve due to their high fermentation rates and spoilage within a few days of production, developing flavours that might reduce the palatability [42]. Thus, we cannot exclude the possibility that the inclusion of the BSWG in the TMR diet had a negative effect on the intake, health, and growth of animals [43].

The ADG was 16% lower in the TMR compared to the Control treatment, in full accordance with lower DMI and ME intake and similar feed efficiency. The ADG presented here are in line with those obtained in a similar trial with Limousine and Retinta young bulls that also compared a concentrate diet with a TMR diet [10]. The slaughter weight, cold carcass weight, LW gain, and carcass gain all follow the same consistent and expected trend of being lower for TMR than for Control, although the power of the experiment was not enough to validate these effects. The ADG was obtained through a random regression method, which is more powerful than the simple mixed model applied to the other variables. We did not find differences in the dressing percentage, which is surprising considering the bulkiness of the TMR diet compared to the Control. In fact, increasing both the DMI and forage proportion in diets is expected to increase the weight of the digestive tract and thus decrease the dressing percentage [7,44]. Probably the higher DMI in animals fed Control compensated for the higher bulkiness of the TMR diet, resulting in a similar dressing percentage.

Despite the decrease in DMI, the inclusion of maize silage in beef cattle diets can be desirable, reducing the amount of concentrate, reducing feed costs, and preventing losses due to rumen acidosis [45]. Among forages, maize silage is notably rich in starch and has a high quality of its fibrous fraction [46]. In fact, the feed costs and feed cost per gain were reduced in our experiment. In Portugal, the compound feed industry is

heavily dependent on imported cereals and oilseed-derived feedstuffs and thus on their international price fluctuations [7]. Partial replacement of concentrate by on-farm or locally produced maize silage warrants an improved predictability of feed costs. Our data are consistent with previous studies of our team [7,47] that found that the costs of feeding diets based on forages and agro-industrial by-products that replace cereals are competitive with commercial concentrates. Nevertheless, our results only relate to the direct costs of feed ingredients, excluding costs associated with feed management such as labour, energy, and equipment.

4.2. Carcass Composition

We estimated the carcass muscle, bone, and fat composition using a classic approach that uses the dissection of the 9th, 10th, and 11th rib sections [18]. Despite genetic and management development of beef production systems, the method continues to be used, and the lean and fat values obtained from the dissection of the 9th, 10th, and 11th rib sections are still strongly correlated with the overall carcass composition, particularly for fat and lean content, with correlation coefficients of 0.87 and 0.80, respectively [48]. This approach allows us to estimate the carcass composition and hence the carcass weight, protein, and energy gain. Naturally, estimations need to be interpreted with caution. In fact, previous studies have suggested that breed differences could explain the differences in the percentage of lean meat, fat, and bone and stated that it was necessary to make small adjustments according to the breed analysed [49,50]. In the present experiment, crossbred Angus × Limousin or Charolais young bulls were used, and in fact, the proportions of lean and fat tissue observed were higher than those reported in dissections of Holstein steers [51]. Duckett et al. [52] reported that crossbred Angus steers fed on pastures presented carcasses with higher lean percentage and less fat than those fed concentrate diets. In our experiment, we did not detect differences between TMR and Control. The variability of estimated carcass fat was large and independent of the treatments, with values ranging from 15.8% to 27.9%. As no effect of treatments on carcass composition was detected, the estimates of CW, energy, and protein gain follow closely that of LW gain.

4.3. Methane Emissions

Individual CH_4 production was measured using a GF unit. This involves the animals voluntarily visiting the equipment during the period when it is available. This variation in the number of records obtained from each animal throughout the trial is due to differences in behaviour towards the equipment and the dominance of some animals over others. In average daily CH_4 production, a high individual variation was observed, suggesting a genetic variability between individuals [53], with average values ranging between 89 and 157 g/d for the Control and between 153 and 234 g/day for the TMR diet. This may be related to differences in rumen microbiota, particularly the abundance of the *Methanobrevibacter* genus, resulting in variations in the expression of methyl-coenzyme M reductase, which catalyses methanogenesis [54]. In fact, several studies have demonstrated correlations between host genotype and CH_4 emissions, which may explain part of the variation observed between individuals [54]. Another possible explanation is the individual feed passage rate, a biological factor that appears to be inversely proportional to individual CH_4 production [55]. The values obtained in the present experiment were consistent with previous studies on beef cattle that obtained CH_4 emissions of 22 g/kg DMI [56] and 20 g/kg DMI [57] for TMR diets, and of 11 g/kg DMI [7] for concentrate diets.

The emission of CH_4 by animals represents a net loss of feed energy, meaning it cannot be utilised by ruminants for productive purposes [58]. Our data indicate that CH_4 energy losses are 3% of GE intake for Control and 5.7% of GE intake for TMR treatment. These

results are consistent with the values reported by Johnson et al. [59]: CH_4 energy losses of around 5.8 to 6.5% of GE intake for all categories and classes of cattle, except for the single situation of a feedlot with high concentrate content, where the typical CH_4 loss can drop to approximately 3%. Thus, as expected, we confirm that a maize silage-based TMR diet, due to its larger contribution of fibre, results in more CH_4 emission and energy losses than a concentrate-based diet.

The GF equipment also provides estimates of animals' CO_2 emissions, despite having no participation in computing the GHG of animals, as the C involved is part of biogenic C recycling, it reflects the oxidation of organic matter and thus the energy yield of the animal [58]. The evaluation of the energy balance by indirect calorimetry also requires information on O_2 consumption, N balance, and retained energy [60]. Nevertheless, the use of the production of CO_2 to estimate energy metabolism and CH_4 in cattle has been explored [61]. The CO_2 produced averaged did not differ between diets, but due to reduced DMI, animals fed TMR produced more CO_2 yield (kg/kg DMI) than those fed the Control diet. Higher CO_2 yields in TMR treatment might be due to lower dietary fat content [62] and putative decreased retained energy. The ratio CH_4:CO_2 (g/kg) is almost double in TMR treatment than in Control, confirming the usefulness of the CH_4:CO_2 ratio as a proxy of relative CH_4 as proposed in the Sniffer methodological approach [62].

4.4. Meat Quality

The quality of meat produced is another important factor that should be taken into consideration when evaluating both feeding systems. We could not detect relevant differences in the composition of meat and its physical quality traits, probably because of a lack of statistical power, but the sensorial analysis clearly pointed to a better eating quality of meat from animals fed TMR compared to those fed Control. In fact, the numerically lower shear force of TMR meat compared to Control is very consistent with the significantly higher score for tenderness given by the panellists. Meat colour is also an important meat quality trait, and consumers value the bright red colour as an indicator of freshness and healthiness [63]. The L* was higher in TMR meat, which indicates slightly higher lightness, and it has also been recognised that for a given b*, the colour acceptability increased linearly with increasing L* [64]. This suggests that the colour of TMR beef would be slightly more acceptable to the consumers. Diet had no significant effect on adipose tissue colour, which was white in both diets, contrary to what has been observed in several studies, which state that the fat of animals fed concentrates is whiter and that of animals fed TMR is more yellowish [41,65,66].

In addition to tenderness, the panellists also score the TMR meat better for juiciness and overall acceptability traits. The higher juiciness for beef from animals fed TMR diets compared to concentrate-based diets was also reported by [10,67].

The health benefits of n − 3 long-chain PUFA are widely recognized and there are official recommendations for the minimal daily intake for adults based on cardiovascular disease risk [68]. Lean meat from forage-fed animals can be a relevant dietary source of n − 3 long-chain PUFA, particularly where the consumption of marine fish and seafood is low [69,70]. Due to the extensive biohydrogenation in the rumen, ruminant meat is also a source of rumenic acid (*c*9,*t*11-18:2), the major conjugated linoleic acid (CLA) isomer, and of its main precursor vaccenic acid (*t*11-18:1). Both biohydrogenation intermediates have numerous health-promoting effects, including anti-carcinogenic properties, as reviewed by Vahmani et al. [70]. One of our hypotheses was that the FA composition of meat tissues would be improved with the incorporation of forage in the diet, particularly that of the n − 3 PUFA, *t*11-18:1, and *c*9,*t*11-18:2. In fact, increasing forages in ruminant diets often led to an increase in n − 3 PUFA on meat, as we observed in a previous experiment when

we compared a concentrated-based finishing diet with a biodiverse haylage-based TMR diet [7]. Moreover, the increased incorporation of forage in ruminants tends to prevent the establishment of *trans*-10 shifted rumen biohydrogenation pathways (hereafter trans-10 shift) and consequent production of the undesirable *t*10-18:1 isomer instead of the healthier *t*11-18:1 and *c*9,*t*11-18:2 [69,71]. In the present experiment, the FA composition of lean meat was very similar between treatments and the expected increase in n − 3 PUFA in meat lipids was not observed, although the meat from TMR-fed animals presented much lower content of 18:2n − 6 and of several BHI, particularly of *t*10-18:1, than in the Control treatment. In the present experiment, the forage used in the TMR was maize silage. When compared to most fresh and preserved forages, maize silage is notable for presenting a low 18:3n − 3 and high 18:2n − 6 content [72]. Thus, the concentration of 18:3n − 3 in the TMR diet compared to Control (38.5 vs. 15.3 g/kg FA) was mild compared to what was found in [7] (152 g/kg FA in TMR vs. 23.3 g/kg in Control). This, together with the lower fat content of the TMR diet and its lower DMI, resulted in approximately similar 18:3n − 3 intake (14 g/d in TMR vs. 11 g/d in Control), explaining the absence of response on n − 3 PUFA in meat.

The FA profile of intermuscular adipose tissue is more informative regarding the dietary effects on lipid metabolism of animals. Although both diets contained high amounts of starch and 18:2n − 6, the Control diet contained more starch (396 vs. 312 g/kg DM) and fat (75 vs. 51 g/kg DM) and less NDF (321 vs. 491 g/kg DM) than the TMR diet. The high-fat intake in the Control treatment is consistent with augmented BHI and reduced de novo synthesis of 16:0 [73]. In ruminants fed pasture/forage-based low-starch diets, commonly, the *t*11-18:1 is the overwhelming BHI formed and thus the *t*10-18:1/*t*11-18:1 ratio is well below 1 [69,71]. However, both high dietary content of starch and 18:2n − 6 favour the establishment of the trans-10 shift in the rumen [71]. In fact, the *t*10-18:1/*t*11-18:1 ratio in adipose tissue was close to 1.1 for TMR and 3.2 for Control. The *trans*-10 shift was much more exacerbated in the Control than in TMR, due to a large increase in *t*10-18:1. Despite the *t*11-18:1 remaining similar in both diets, the c9,t11-18:2 was higher in the Control. The *c*9,*t*11-18:2 present in the tissues is mainly derived from the endogenous desaturation of *t*11-18:1, which originates in the rumen, by the stearoyl-CoA desaturase (SCD) [74]. Concentrate-fed ruminants showed higher SCD activity in ImF, increasing the conversion of *t*11-18:1 into *c*9,*t*11-18:2 [69]. This probably explains the increased *c*9,*t*11-18:2, as well as other SCD products like *c*9-18:1, in Control and is also consistent with augmented SCD-17, a proxy of SCD activity [69]. Moreover, the contribution of conjugated isomer *t*7,*c*9-18:2 to the reported increased c9,t11-18:2 cannot be excluded, as in our chromatographic conditions, both isomers coelute. The *t*7,*c*9-18:2 is also formed by SCD through the desaturation of t7-18:1 and is commonly increased in ruminants with concentrate feeding [75].

Overall, the FA profile of both lean and fat meat tissues of TMR-fed animals was only slightly improved, through the mitigation of the *t*10-shift, lower trans-MUFA, and lower n − 3/n − 6 PUFA ratio. Despite that, the PUFA content was reduced, mostly due to diminished 18:2n − 6 content as the supply of n − 3 PUFA remained quite low in both diets, below 10 mg/100 g of lean meat.

4.5. Carbon Footprint

The evaluation of GHG emissions of a feeding system should not be based only on direct digestive tract emissions from animals, and other GHG sources must be considered using an LCA perspective. To address this, we calculated the overall CFp over the 12-week test period only, as the contribution of the other production and consumption stages is common to both treatments. Most of the emission sources were calculated using emission factors, except for rumen emissions, which were measured. It is important to note that the

assessment only involved young bulls. It did not consider any pre-test emissions, such as those from the breeding herd and calves prior to reaching the finishing stage, nor those from the dam. Moreover, as this earlier cow-calf stage has a large contribution in the whole production cycle (accounting for about 80% of total emissions) [12,76,77], there would be a dilution of the finishing stage and low sensitivity of the analysis to detect the effect under study. The CFp results were expressed using the LW gain, CW gain, and CPr gain as functional units. The LCA results show that the large differences observed for digestive CH_4 emissions between treatments were compensated for by the also large differences of GHG emissions attributed to the feeds and to a lesser extent of manure produced. Thus, a similar CFp was obtained for both treatments (≈6.0 kg CO_2eq/kg LW gain, ≈9.8 kg CO_2eq/kg CW gain, and ≈51.6 kg CO_2eq/kg CPr gain). These confirm the previous results obtained by our team, where Limousine × Alentejana crossbred young bulls were fed for 64 days either a concentrate-based diet or a biodiverse forage-based TMR, and the CFp was 6.6 kg CO_2eq/kg LW gain for both treatments [7]. Berton et al. [78] reported a higher value of 8.3 kg CO_2eq/kg LW gain for bulls of French breeds and crosses. In addition to the differences in animal genetics and feeding management, these authors used published databases to estimate the rumen emissions, whereas we used individual data on rumen digestive emissions and intake, as well as detailed feeding information.

The CFp results clearly demonstrate the trade-off between concentrate-based diets and forage-based diets in beef finishing, as the GHG contribution of the compound feed production chain can be large enough to compensate for the larger digestive tract CH_4 yield of forage-based diets.

5. Conclusions

The crossbred young bulls finished with a TMR diet presented less 13% DMI and 15% of ADG and produced 59% more daily CH_4 compared to those fed the Control diet. However, when assessed through a partial LCA, these differences were fully offset by the lower feed costs and lower GHG emissions associated with the lower compound feed incorporation in diets. The carcass composition remained unaltered, but the eating quality of the meat from bulls finished with a TMR diet improved, showing better tenderness, juiciness, and overall acceptability scores. The FA composition of lean meat was fairly similar between diets, but with less *t*10-18:1 and hence total trans-MUFA, which is an improvement regarding the consumer's health.

These results point out that replacing conventional concentrate-based diets with maize silage-based TMR diets could improve the sustainability of beef cattle finishing systems commonly used in south-western Europe, by decreasing costs and reducing reliance on imported grains and off-farm inputs, while maintaining CFp and improving beef quality. This offers a practical pathway for aligning intensive beef production with regional and EU sustainability goals.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su17188417/s1, Table S1: Effects of the diet on fatty acid (mg/100 g of meat) of *m. Longissimus lumborum* (LL) of crossbred Angus young bulls. Table S2: Effects of the diet on fatty acid (FA) (mg/g of total FA) of intermuscular adipose tissue from *m. Longissimus lumborum* (LL) of crossbred Angus young bulls.

Author Contributions: Conceptualization, T.D. and R.J.B.B.; formal analysis, D.M.S., N.R., R.F.M.T., S.P.A. and R.J.B.B.; investigation, D.M.S., S.B., J.M.A. and R.J.B.B.; resources, T.D.; data curation, I.G.; writing—original draft preparation, D.M.S. and R.J.B.B.; writing—review and editing, D.M.S., S.B., N.R., I.G., J.M.A., R.F.M.T., J.S.-S., S.P.A., T.D., C.M., G.M.M. and R.J.B.B.; supervision, C.M., G.M.M. and R.J.B.B.; project administration, T.D. and R.J.B.B.; funding acquisition, T.D. and R.J.B.B. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by a Ph.D. research grant awarded by Fundação para a Ciência e Tecnologia (FCT) to D.M. Soares (2022.13385.BDANA) and by the FCT-funded projects UIDB/00276/2020 (CIISA), UIDB/50009/2025, UIDP/50009/2025 and LA/P/0083/2020 (MARETEC) and LA/P/0059/2020 (AL4AnimalS). Funding from projects GreenBeef (POCI-01-0247-FEDER-047050/LISBOA-01-FEDER-047050), AVALON (COMPETE2030-FEDER-02288900) and CYTED, Red SISPEC (125RT0167—Linha 1 Agroalimentación) is also acknowledged.

Institutional Review Board Statement: The animal study protocol was approved by the Research and Education Ethics Committee of the Faculty of Veterinary Medicine, University of Lisbon, Portugal (Proc. 038/2023).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Data will be made available on request.

Acknowledgments: The authors would like to thank the staff of the Terraprima—Sociedade Agrícola (Quinta da França, Portugal) for their support in daily activities related to the management of the animals and the processing of samples, and the Best Farmer staff, particularly José Fraga, Francisca Manarte and Luísa Silva, for their support in choosing the animals used in the present study.

Conflicts of Interest: Author Nuno Rodrigues and Ivo Gama were employed by the company Terraprima-Serviços Ambientais. Author Sílvia Bernardino was employed by the company Terraprima—Sociedade Agrícola. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. FAO. *Pathways Towards Lower Emissions—A Global Assessment of the Greenhouse Gas Emissions and Mitigation Options from Livestock Agrifood Systems*; FAO: Rome, Italy, 2023. [CrossRef]
2. Steinfeld, H.; Gerber, P.; Wassenaar, T.D.; Castel, V.; Rosales, M.; de Haan, C. *Livestock's Long Shadow: Environmental Issues and Options*; FAO: Rome, Italy, 2006; Available online: https://www.fao.org/3/a0701e/a0701e00.htm (accessed on 4 February 2025).
3. Tedeschi, L.O.; Beauchemin, K.A. Galyean appreciation club review: A holistic perspective of the societal relevance of beef production and its impacts on climate change. *J. Anim. Sci.* **2023**, *101*, skad024. [CrossRef]
4. Blaustein-Rejto, D.; Soltis, N.; Blomqvist, L. Carbon opportunity cost increases carbon footprint advantage of grain-finished beef. *PLoS ONE* **2023**, *18*, e0295035. [CrossRef]
5. Haque, M.N. Dietary manipulation: A sustainable way to mitigate methane emissions from ruminants. *J. Anim. Sci. Technol.* **2018**, *60*, 15. [CrossRef]
6. Hristov, A.N.; Oh, J.; Giallongo, F.; Frederick, T.W.; Harper, M.T.; Weeks, H.L.; Branco, A.F.; Moate, P.J.; Deighton, M.H.; Williams, S.R.O.; et al. An inhibitor persistently decreased enteric methane emission from dairy cows with no negative effect on milk production. *Proc. Natl. Acad. Sci. USA* **2015**, *112*, 10663–10668. [CrossRef]
7. Santos-Silva, J.; Alves, S.P.; Francisco, A.; Portugal, A.P.; Dentinho, M.T.; Almeida, J.; Da Silva, J.L.R.; Fialho, L.; Cachucho, L.; Jerónimo, E.; et al. Forage based diet as an alternative to a high concentrate diet for finishing young bulls—Effects on growth performance, greenhouse gas emissions and meat quality. *Meat Sci.* **2023**, *198*, 109098. [CrossRef]
8. Instituto Nacional de Estatística (INE). Estatísticas Agrícolas 2020 [Dataset]. 2021. Available online: https://www.ine.pt/xurl/pub/437147278 (accessed on 5 January 2025).
9. Keady, T.W.J.; Lively, F.O.; Kilpatrick, D.J.; Moss, B.W. Effects of replacing grass silage with either maize or whole-crop wheat silages on the performance and meat quality of beef cattle offered two levels of concentrates. *Anim. Int. J. Anim. Biosci.* **2007**, *1*, 613–623. [CrossRef] [PubMed]
10. Avilés, C.; Martínez, A.L.; Domenech, V.; Peña, F. Effect of feeding system and breed on growth performance, and carcass and meat quality traits in two continental beef breeds. *Meat Sci.* **2015**, *107*, 94–103. [CrossRef] [PubMed]
11. Beauchemin, K.A.; Ungerfeld, E.M.; Abdalla, A.L.; Alvarez, C.; Arndt, C.; Becquet, P.; Benchaar, C.; Berndt, A.; Mauricio, R.M.; McAllister, T.A.; et al. Invited review: Current enteric methane mitigation options. *J. Dairy Sci.* **2022**, *105*, 9297–9326. [CrossRef]
12. Beauchemin, K.A.; Henry Janzen, H.; Little, S.M.; McAllister, T.A.; McGinn, S.M. Life cycle assessment of greenhouse gas emissions from beef production in western Canada: A case study. *Agric. Syst.* **2010**, *103*, 371–379. [CrossRef]
13. European Union. Directive 2010/63/EU of the European Parliament and of the Council of 22 September 2010 on the protection of animals used for scientific purposes. *Off. J. Eur. Union L* **2010**, *276*, 33–79.

14. Sauvant, D.; Perez, J.-M.; Tran, G. (Eds.) *Tables of Composition and Nutritional Value of feed Materials: Pigs, Poultry, Cattle, Sheep, Goats, Rabbits, Horses and Fish*; Brill, Wageningen Academic: Leiden, The Netherlands, 2004. [CrossRef]
15. European Union Council regulation (EC) nº 1099/2009 of 24 September 2009 on the protection of animals at the time of killing. *Off. J. Eur. Union L* **2009**, *303*, 1–30.
16. Commission Regulation (EC). Commission Regulation (EC) No 1249/2008 of 10 December 2008. Official Journal of the European Union. 2008. Available online: http://data.europa.eu/eli/reg/2008/1249/oj (accessed on 16 September 2025).
17. Institutional Meat Purchase Specifications (IMPS). *Fresh Beef Series 100*; United States Department of Agriculture: Washington, DC, USA, 2014.
18. Hankins, O.G.; Howe, P.E. *Estimation of the Composition of Beef Carcasses and Cuts*; (U.S. Department of Agriculture Technical Bulletin No. 926); U.S. Department of Agriculture: Washington, DC, USA, 1946.
19. Agricultural Research Council. *The Nutrient Requirements of Ruminant Livestock: Technical Review by an Agricultural Research Council Working Party*; Commonwealth Agricultural Bureaux: Wallingford, UK, 1980.
20. Clegg, K.M. The application of the anthrone reagent to the estimation of starch in cereals. *J. Sci. Food Agric.* **1956**, *7*, 40–44. [CrossRef]
21. Van Soest, P.J.; Robertson, J.B.; Lewis, B.A. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* **1991**, *74*, 3583–3597. [CrossRef]
22. Tilley, J.M.A.; Terry, R.A. A two-stage technique for the in vitro digestion of forage crops. *Grass Forage Sci.* **1963**, *18*, 104–111. [CrossRef]
23. Sukhija, P.S.; Palmquist, D.L. Rapid method for determination of total fatty acid content and composition of feedstuffs and feces. *J. Agric. Food Chem.* **1988**, *36*, 1202–1206. [CrossRef]
24. *ISO 1442:1997*; Meat and Meat Products—Determination of Moisture Content. International Organization for Standardization: Geneva, Switzerland, 1997.
25. *NP 1614*; Norma Portuguesa NP 1614 2002: Carne e produtos cárneos: Determinação do teor de humidade método de referência. Instituto Português de Qualidade: Caparica, Portugal, 2002.
26. AOAC 920.39. 4.5.01 AOAC Official Method 920.39 Fat (Crude) or Ether Extract in Animal Feed. 1920. Available online: http://138.128.182.170/~aoxlabsgc/files/NORM-TC/005%20Grasa/Norma%20tecnica%20del%20analisis%20de%20grasa.pdf (accessed on 15 January 2025).
27. *NP 3441*; Norma Portuguesa NP 3441 1990: Carnes, Derivados e Produtos Cárneos Determinação do pH Processo de Referência. Instituto Português de Qualidade: Caparica, Portugal, 2007.
28. Folch, J.; Lees, M.; Stanley, G.H.S. A simple method for the isolation and purification of total lipides from animal tissues. *J. Biol. Chem.* **1957**, *226*, 497–509. [CrossRef] [PubMed]
29. Alves, S.P.; Raundrup, K.; Cabo, Â.; Bessa, R.J.B.; Almeida, A.M. Fatty acid composition of muscle, adipose tissue and liver from muskoxen (*Ovibos moschatus*) living in west Greenland. *PLoS ONE* **2015**, *10*, e0145241. [CrossRef]
30. Alves, S.P.; Bessa, R.J.B. Comparison of two gas–liquid chromatograph columns for the analysis of fatty acids in ruminant meat. *J. Chromatogr. A* **2009**, *1216*, 5130–5139. [CrossRef]
31. Vahmani, P.; Rolland, D.C.; Gzyl, K.E.; Dugan, M.E.R. Non-conjugated cis/trans 18:2 in beef fat are mainly δ-9 desaturation products of trans-18:1 isomers. *Lipids* **2016**, *51*, 1427–1433. [CrossRef]
32. American Meat Science Association (AMSA). *AMSA Meat Color Measurement Guidelines*; American Meat Science Association: Kearney, MO, USA, 2012; pp. 1–135.
33. Cross, H.R.; Moen, R.M.; Stanfield, M.S. Training and testing of judges for sensory analysis of meat quality. *Food Technol.* **1978**, *32*, 48–54.
34. *ISO 14040:2006*; Environmental Management—Life Cycle Assessment—Principles and Framework. International Organization for Standardization: Geneva, Switzerland, 2006. Available online: https://www.iso.org/standard/37456.html (accessed on 6 January 2025).
35. IPCC. *Climate Change 2007: Synthesis Report: A Report of the Intergovernmental Panel on Climate Change*; IPCC: Geneva, Switzerland, 2007.
36. IPCC. *Intergovernmental Panel on Climate Change. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*; IPCC: Geneva, Switzerland, 2006.
37. NIR. Portuguese National Inventory Report on Greenhouse Gases, 1990–2019. 2021. Available online: https://unfccc.int/documents/271508 (accessed on 16 September 2025).
38. ECOALIM. An LCA Database for Animal Feed. 2014. Available online: https://eng-ecoalim.hub.inrae.fr/ (accessed on 6 January 2025).
39. CarbonCloud Database. Explore Climate Footprints. Available online: https://apps.carboncloud.com/climatehub/ (accessed on 6 January 2025).
40. Forbes, J.M. *Voluntary Food Intake and Diet Selection in Farm Animals*, 2nd ed.; CABI: Wallingford, UK, 2007.

41. Cooke, D.; Monahan, F.; Brophy, P.; Boland, M. Comparison of concentrates or concentrates plus forages in a total mixed ration or discrete ingredient format: Effects on beef production parameters and on beef composition, colour, texture and fatty acid profile. *Ir. J. Agric. Food Res.* **2004**, *43*, 201–216.
42. Johnson, C.O.L.E.; Huber, J.T.; King, K.J. Storage and utilization of brewers wet grains in diets for lactating dairy cows. *J. Dairy Sci.* **1987**, *70*, 98–107. [CrossRef]
43. Valverde, P. Barley spent grain and its future. *Cerveza Malta* **1994**, *22*, 7–26.
44. Zinn, R.A.; Barreras, A.; Owens, F.N.; Plascencia, A. Performance by feedlot steers and heifers: Daily gain, mature body weight, dry matter intake, and dietary energetics. *J. Anim. Sci.* **2008**, *86*, 2680–2689. [CrossRef]
45. Martin, C.; Morgavi, D.; Doreau, M. Methane mitigation in ruminants: From microbe to the farm scale. *Animal* **2010**, *4*, 351–365. [CrossRef]
46. Allen, V.G.; Fontenot, J.P.; Kelly, R.F.; Notter, D.R. Forage systems for beef production from conception to slaughter: III. Finishing systems. *J. Anim. Sci.* **1996**, *74*, 625–638. [CrossRef]
47. Santos-Silva, J.; Francisco, A.; Portugal, A.P.; Paulos, K.; Dentinho, M.T.; Almeida, J.M.; Regedor, L.; Fialho, L.; Cachucho, L.; Jerónimo, E.; et al. Effects of partial substitution of grain by agroindustrial byproducts and sunflower seed supplementation in beef haylage-based finisher diets on growth, in vitro methane production and carcass and meat quality. *Meat Sci.* **2022**, *188*, 108782. [CrossRef] [PubMed]
48. Cooper, K.B.; Pillmore, S.L.; Kirkpatrick, T.J.; Wesley, K.R.; Lawrence, T.; Tennant, T.; Nichols, W.; Hutcheson, J. PSI-5 Comparison of ninth-tenth-eleventh rib section to carcass components in serially harvested steer carcasses. *J. Anim. Sci.* **2020**, *98* (Suppl. 4), 273–274. [CrossRef]
49. Koch, R.M.; Dikeman, M.E.; Lipsey, R.J.; Allen, D.M.; Crouse, J.D. Characterization of biological types of cattle—Cycle ii: iii. Carcass composition, quality and palatability. *J. Anim. Sci.* **1979**, *49*, 448–460. [CrossRef]
50. Nour, A.Y.M.; Thonney, M.L. Technical note: Chemical composition of Angus and Holstein carcasses predicted from rib section composition. *J. Anim. Sci.* **1994**, *72*, 1239–1241. [CrossRef]
51. McEvers, T.J.; May, N.D.; Reed, J.A.; Walter, L.J.; Hutcheson, J.P.; Lawrence, T.E. Estimation of carcass composition using rib dissection of calf-fed Holstein steers supplemented zilpaterol hydrochloride. *J. Anim. Sci.* **2018**, *96*, 1396–1404. [CrossRef]
52. Duckett, S.K.; Neel, J.P.S.; Sonon, R.N., Jr.; Fontenot, J.P.; Clapham, W.M.; Scaglia, G. Effects of winter stocker growth rate and finishing system on: II. Ninth–tenth–eleventh-rib composition, muscle color, and palatability1. *J. Anim. Sci.* **2007**, *85*, 2691–2698. [CrossRef]
53. Black, J.L.; Davison, T.M.; Box, I. Methane emissions from ruminants in Australia: Mitigation potential and applicability of mitigation strategies. *Animals* **2021**, *11*, 951. [CrossRef]
54. Abbott, D.W.; Aasen, I.M.; Beauchemin, K.A.; Grondahl, F.; Gruninger, R.; Hayes, M.; Huws, S.; Kenny, D.A.; Krizsan, S.J.; Kirwan, S.F.; et al. Seaweed and seaweed bioactives for mitigation of enteric methane: Challenges and opportunities. *Animals* **2020**, *10*, 2432. [CrossRef]
55. Goopy, J.P.; Donaldson, A.; Hegarty, R.; Vercoe, P.E.; Haynes, F.; Barnett, M.; Oddy, V.H. Low-methane yield sheep have smaller rumens and shorter rumen retention time. *Br. J. Nutr.* **2014**, *111*, 578–585. [CrossRef] [PubMed]
56. Smith, P.E.; Waters, S.M.; Kenny, D.A.; Kirwan, S.F.; Conroy, S.; Kelly, A.K. Effect of divergence in residual methane emissions on feed intake and efficiency, growth and carcass performance, and indices of rumen fermentation and methane emissions in finishing beef cattle. *J. Anim. Sci.* **2021**, *99*, skab275. [CrossRef]
57. Ryan, C.V.; Pabiou, T.; Purfield, D.C.; Conroy, S.; Kirwan, S.F.; Crowley, J.J.; Murphy, C.P.; Evans, R.D. Phenotypic relationship and repeatability of methane emissions and performance traits in beef cattle using a GreenFeed system. *J. Anim. Sci.* **2022**, *100*, skac349. [CrossRef] [PubMed]
58. Blaxter, K.L. *The Energy Metabolism of Ruminants*; London Hutchinson: London, UK, 1967.
59. Johnson, D.E.; Hill, T.M.; Carmean, B.R.; Branine, M.E.; Lodman, D.W.; Ward, G.M. *Perspectives on Ruminant Methane Emission*; Beef Program Report; Department of Animal Sciences, Colorado State University: Fort Collins, CO, USA, 1993.
60. McDonald, P.; Greenhalgh, J.F.D.; Morgan, C.; Edwards, R.; Sinclair, L.; Wilkinson, R. *Animal Nutrition*; Pearson Education: London, UK, 2021.
61. Kjeldsen, M.H.; Johansen, M.; Weisbjerg, M.R.; Hellwing, A.L.F.; Bannink, A.; Colombini, S.; Crompton, L.; Dijkstra, J.; Eugène, M.; Guinguina, A.; et al. Predicting CO_2 production of lactating dairy cows from animal, dietary, and production traits using an international dataset. *J. Dairy Sci.* **2024**, *107*, 6771–6784. [CrossRef]
62. Madsen, J.; Bjerg, B.S.; Hvelplund, T.; Weisbjerg, M.R.; Lund, P. Methane and carbon dioxide ratio in excreted air for quantification of the methane production from ruminants. *Livest. Sci.* **2010**, *129*, 223–227. [CrossRef]
63. Mancini, R.A.; Hunt, M.C. Current research in meat color. *Meat Sci.* **2005**, *71*, 100–121. [CrossRef]
64. Holman, B.W.B.; Mao, Y.; Coombs, C.E.O.; van de Ven, R.J.; Hopkins, D.L. Relationship between colorimetric (instrumental) evaluation and consumer-defined beef colour acceptability. *Meat Sci.* **2016**, *121*, 104–106. [CrossRef]

65. Bidner, T.D.; Schupp, A.R.; Montgomery, R.E.; Carpenter, J.C., Jr. Acceptability of beef finished on all-forage, forage-plus-grain or high energy diets. *J. Anim. Sci.* **1981**, *53*, 1181–1187. [CrossRef]
66. Schmutz, M.; Weindl, P.; Carrasco, S.; Bellof, G.; Schmidt, E. Effect of breed, grazing system and concentrate supplementation on fattening performance, carcass value and meat quality of steers. *Arch. Anim. Breed.* **2013**, *56*, 943–957. [CrossRef]
67. Kim, T.-I.; Mayakrishnan, V.; Lim, D.-H.; Yeon, J.-H.; Baek, K.-S. Effect of fermented total mixed rations on the growth performance, carcass and meat quality characteristics of Hanwoo steers. *Anim. Sci. J.* **2018**, *89*, 606–615. [CrossRef]
68. European Food Safety Authority. *Annual Report 2012*; EFSA: Parma, Italy, 2012.
69. Bessa, R.J.B.; Alves, S.P.; Santos-Silva, J. Constraints and potentials for the nutritional modulation of the fatty acid composition of ruminant meat. *Eur. J. Lipid Sci. Technol.* **2015**, *117*, 1325–1344. [CrossRef]
70. Vahmani, P.; Ponnampalam, E.N.; Kraft, J.; Mapiye, C.; Bermingham, E.N.; Watkins, P.J.; Proctor, S.D.; Dugan, M.E.R. Bioactivity and health effects of ruminant meat lipids: Invited review. *Meat Sci.* **2020**, *165*, 108114. [CrossRef] [PubMed]
71. Alves, S.P.; Vahmani, P.; Mapiye, C.; McAllister, T.A.; Bessa, R.J.B.; Dugan, M.E.R. Trans-10 18:1 in ruminant meats: A review. *Lipids* **2021**, *56*, 539–562. [CrossRef]
72. Glasser, F.; Doreau, M.; Maxin, G.; Baumont, R. Fat and fatty acid content and composition of forages: A meta-analysis. *Anim. Feed Sci. Technol.* **2013**, *185*, 19–34. [CrossRef]
73. Bessa, R.J.B.; Portugal, P.V.; Mendes, I.A.; Santos-Silva, J. Effect of lipid supplementation on growth performance, carcass and meat quality and fatty acid composition of intramuscular lipids of lambs fed dehydrated lucerne or concentrate. *Livest. Prod. Sci.* **2005**, *96*, 185–194. [CrossRef]
74. Griinari, J.M.; Bauman, D.E. Biosynthesis of conjugated linoleic acid and its incorporation into meat and milk in ruminants. In *Advances in Conjugated Linoleic Acid Research*; Yurawecz, M.P., Mossoba, M.M., Kramer, J.K.G., Pariza, M.W., Nelson, G.J., Eds.; AOCS Press: Champaign, IL, USA, 2019.
75. Aldai, N.; de Renobales, M.; Barron, L.J.R.; Kramer, J.K.G. What are the trans fatty acids issues in foods after discontinuation of industrially produced trans fats? Ruminant products, vegetable oils, and synthetic supplements. *Eur. J. Lipid Sci. Technol.* **2013**, *115*, 1378–1401. [CrossRef]
76. O'Brian, D.; Shalloo, L. *A Review of Livestock Methane Emission Factors (2016-CCRP-DS.11) EPA Research Report*; Environmental Protection Agency. Johnstown Castle, Co.: Wexford, Ireland, 2021.
77. Rotz, C.A.; Asem-Hiablie, S.; Dillon, J.; Bonifacio, H. Cradle-to-farm gate environmental footprints of beef cattle production in Kansas, Oklahoma, and Texas. *J. Anim. Sci.* **2015**, *93*, 2509–2519. [CrossRef] [PubMed]
78. Berton, M.; Cesaro, G.; Gallo, L.; Pirlo, G.; Ramanzin, M.; Tagliapietra, F.; Sturaro, E. Environmental impact of a cereal-based intensive beef fattening system according to a partial life cycle assessment approach. *Livest. Sci.* **2016**, *190*, 81–88. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.